

Supermarket Sales Analysis

AICTE | IBM SkillsBuild Data Analytics with AI Internship 2026

1. Problem Statement

Analyze supermarket sales transactions and identify useful information about products, branches, categories, customers, payment methods and customer ratings. The objective is to convert transaction data into simple business insights that can support inventory, branch and customer-service decisions.

2. Dataset

The supplied dataset contains 500 transactions and 13 fields: Invoice ID, Date, Branch, City, Customer Type, Gender, Product, Category, Quantity, Unit Price, Payment, Rating, Sales. Sales values were checked against $\text{Quantity} \times \text{Unit Price}$.

3. Methodology

- Load and inspect the transaction dataset.
- Check missing values and convert date/numeric fields to suitable types.
- Validate sales using $\text{Quantity} \times \text{Unit Price}$.
- Group transactions by product, branch, category, payment method and customer type.
- Calculate totals, averages and transaction counts.
- Create charts to communicate the results.
- Translate the observed patterns into business recommendations.

4. Key Findings

Metric	Result
Total transactions	500
Total sales	■244,411.08
Average transaction	■488.82
Highest-sales product	Cheese — ■27,906.30
Highest-sales branch	Branch C (Mumbai) — ■72,469.45
Highest-sales category	Beverages — ■56,108.24
Most-used payment method	UPI — 127 transactions
Average customer rating	3.99/5

5. Business Recommendations

- Monitor stock and promotions for high-sales products, especially Cheese.
- Study the operational and sales patterns of Branch C in Mumbai.
- Continue supporting the frequently used UPI payment method while maintaining alternatives.
- Use customer-rating patterns to identify opportunities for improving service quality.

- Use customer-type spending patterns when designing membership or promotional offers.

6. Conclusion

The analysis demonstrates how transaction-level supermarket data can be transformed into clear business insights using Python and data analytics techniques. The resulting findings can support decisions related to inventory, branch performance, payments and customer experience.